

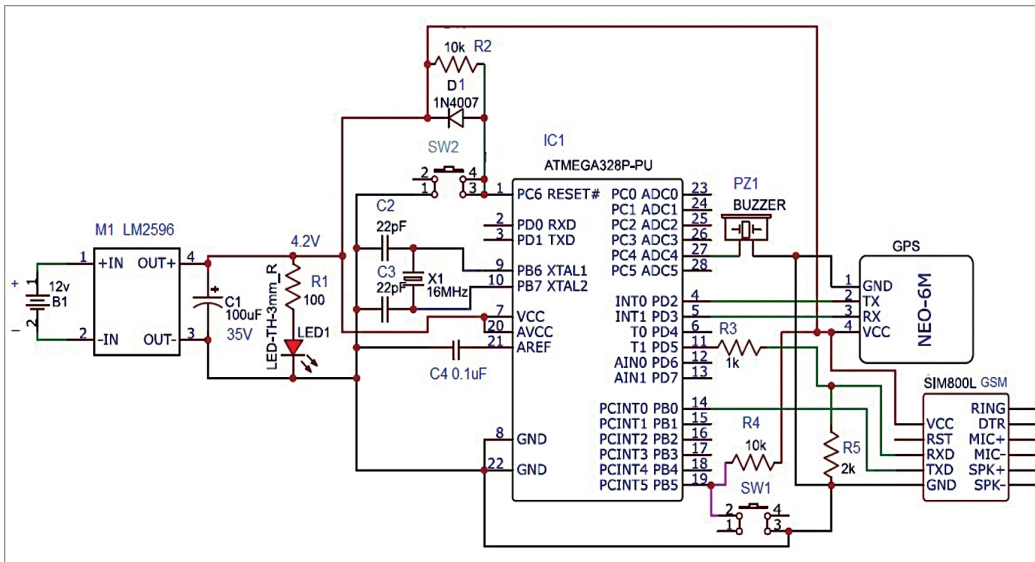


Fig. 6: Circuit diagram

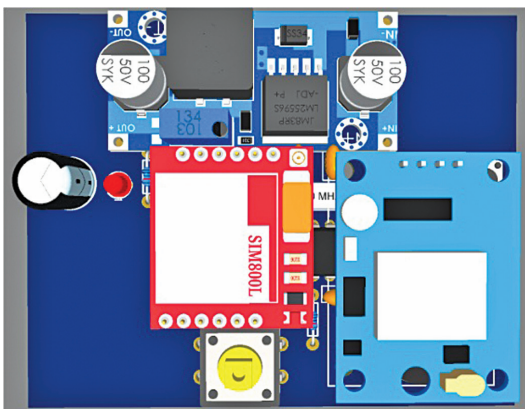


Fig. 7: A 3D view of the prototype

Pin details of SIM800L are shown in Fig. 2.

Buck converter (or step-down converter) is a DC to DC power converter, which steps down voltage (while stepping up current) from its input (supply) to its output (load). A DC-DC step-down converter basically converts a higher input voltage to a lower output voltage by ‘chopping’ it up by rapidly switching the output power. The buck converter module is shown in Fig. 3.

Neo-6M GPS module is a well-performing complete GPS receiver with a built-in 25x25x4mm ceramic antenna, which provides a strong satellite search capability. With the power and signal indicators the

status of the module can be monitored. The GPS module is shown in Fig. 4.

Atmega328P is the microcontroller used in an Arduino board. After uploading the code into this microcontroller using an Arduino board, it can be plugged out and used in the breadboard prototype.

As shown in the block diagram (Fig. 5), there are also some other components like push buttons

(SW1 and SW2), GPS module, GSM module, and power supply. The circuit diagram of the project is shown in Fig. 6. The microcontroller is, of course, the heart of the project.

Whenever the push button SW1 is pressed by the caller, the microcontroller gives the information about the caller, like his/her location and emergency SMS, to the registered number in GSM module. As a result, the receiver mobile receives caller's

SOS message.

As shown in the circuit diagram, push button SW2 is used to reset the circuit (whenever there is something wrong in the program). LED1 is used to indicate the presence of power in the circuit. The circuit can be powered by a 4.2V supply from the converter LM2596 module.

A 3D view of the prototype generated by the software is shown in Fig. 7 and

the author's prototype is shown in Fig. 8.

Software

Arduino IDE 1.8.15 is used to write, compile, and upload the code (smart_SOS_final.ino) into the Arduino board. The code requires TinyGPS++.h header file. The header file can be downloaded from <http://arduinoiana.org/libraries/tinygpsplus/>

Before uploading the code, enter a valid 10-digit phone number in the code

String number = "xxxxxxxxxx";
where 'xxxxxxxxxx' is the phone number to receive the SOS message in the receiver's mobile phone.

Construction and testing

EasyEDA is a free web based EDA (electronics design automation) tool, which gives facility to design schematic diagrams as well as PCB layouts. After completing schematic diagram on EasyEDA platform, select Design→Convert schematic to PCB. Adjust all the components of the circuit in best minimum space. Then select Route→Auto route.

EasyEDA can generate 2D and 3D visualisations of the PCB. To get them, select 2D View and 3D View,

EFY Note

The source code of this project is available

for free download at www.electronicsforu.com